Daniel Taejoon Chong

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Ph.D. Candidate specializing in automated vulnerability discovery and binary reverse engineering. Proven track record discovering hundreds of bugs across critical platforms including PHP, Python, and Ruby through novel vulnerability discovery frameworks. Experienced in developing custom binary emulators and graph-based abstractions for complex system modeling.

EDUCATION

Ph.D. in Computer Science and Engineering (GPA: 4.0/4.0) Anticipated: Spring 2027
College of Engineering and Computer Science, Wright State University Dayton, OH

Master of Science in Computer Engineering (GPA: 4.0/4.0) Spring 2022
College of Engineering and Computer Science, Wright State University Dayton, OH

Bachelor of Science in Computer Engineering (GPA: 3.93/4.0) Spring 2021
College of Engineering and Computer Science, Wright State University Dayton, OH

Bachelor of Science in Computer Science (GPA: 3.93/4.0) Spring 2021
College of Engineering and Computer Science, Wright State University Dayton, OH

RESEARCH EXPERIENCE

Graduate Student **May 2021 – Present**
Wright State University
Department of Computer Science and Engineering
Advisor: Junjie Zhang

Project: Language-agnostic Taxonomy for Variant Analysis: 23 bugs

- Designed an automated taxonomy generation pipeline that abstracts and clusters bug reports into an implementation-neutral taxonomy.

- Implemented a variant analysis pipeline to use taxonomies to guide agentic vulnerability search.

Project: Fuzzing the PHP JIT Optimizer with Cross-Language Structural Transfer: 20 bugs
- Designed a system to extract optimizer-triggering constraints from historical optimizer bugs.
- Developed a statement-labeling system to match statement types to optimizer constraint types.
- Created OptFuzz, a system which fuzzes the PHP optimizer with constraint satisfying seeds.

Project: Fuzzing PHP with an Intermediate Representation: 68 bugs
- Developed SeqIR, a semantics-aware intermediate representation which captures the control and data flows of PHP as hierarchical sequences of nodes.
- Implemented SeqFuzz, a system which mutates and lifts SeqIR to fuzz the PHP interpreter.

Project: Fuzzing PHP with JavaScript PoCs: 50 bugs
- Developed an LLM-based tool to translate JavaScript PoC structures to PHP.
- Developed a novel energy scheduler to increase efficiency of LLM-based mutations.
- Integrated components into TransFuzz, a system which mutates JS PoCs to fuzz PHP.

Project: Reverse Engineering x86 for Function Name Recovery
- Developed an AST-like representation from Ghidra's PCode to support function name recovery from stripped binaries.

Reverse Engineer **October 2022 – October 2023**
Tenet3 Cybernetics

Project: Hardening of Cyber-physical Systems
- Developed novel graph-based binary abstractions for ML-based vulnerability matching.
- Reverse engineered ARM, x86, and MIPS binaries to support cyber-physical system supply chain security efforts.
- Developed LLVM-based program mutator to alter binary form while maintaining functionality.
- Planned and wrote proposals for project funding rounds.

Project: Zero-day Bug Hunting and CTF
- Led company team in reverse engineering capture-the-flag challenges.
- Participated in zero-day bug hunting and exploit generation to evaluate tool effectiveness.

Autonomy Technology Research Center (ATRC) Intern **May 2021 – July 2021**
Air Force Research Laboratory (AFRL)
- Symbolic execution for firmware binary security analysis.
- Control-flow graph recovery of firmware binaries.

Undergraduate Research Assistant **September 2019 – April 2021**
Wright State University
Department of Computer Science and Engineering

Advisor: Junjie Zhang
Sponsor: NHTSA

Project: Vehicle Firmware Security
- Developed a Unicorn-based, ARM 32-bit stripped binary emulator.
- Created a peripheral emulator to simulate peripheral device input into the emulator.
- Wrote a novel algorithm to calculate the loading point of stripped ARM32 binaries.
- Orchestrated components with symbolic execution to create a high-coverage binary fuzzer.

TEACHING EXPERIENCE

Graduate Teaching Assistant **Fall 2023, Spring 2024**
Course: Trust Networks
Responsibilities: Office hours, Grading

Graduate Teaching Assistant **Fall 2022**
Course: Introduction to Quantum Computing
Responsibilities: Supplemental research discussions, Office hours, Grading

HONORS

WSU 2025 College of Engineering and Computer Science Top Student Award

WSU 2024 Computer Science and Engineering Graduate Student Excellence Award

GRANTS AND CONTRACTS

Co-author, USAF SBIR Phase II (AF221-0033), "DARWIN Workbench," Tenet3, LLC.
$1,299,997 (2024)

PUBLICATIONS

"TaxoShop: Taxonomy-Guided Variant Analysis Across Programming Languages". To be submitted. **Daniel Chong**, Paschal Amusuo, Junjie Zhang, Zhiqiang Lin.

"JotFuzz: Fuzzing the PHP JIT Optimizer by Satisfying Optimization Preconditions". Submitted to NDSS 2027. **Daniel Chong**, Junjie Zhang, Zhiqiang Lin.

"TransFuzz: Cross-Language Fuzzing of PHP Interpreters via LLM-Translated JavaScript PoCs." Submitted to USENIX 2027. **Daniel Chong**, Junjie Zhang, Zhiqiang Lin.

"SeqFuzz: Fuzzing the PHP Interpreter with a Semantics-Aware Intermediate Representation." Submitted to FSE 2027. **Daniel Chong**, Junjie Zhang, Zhiqiang Lin.

"FedPDFGuard: A Federated Learning Approach for Robust Detection of Malicious PDF Files." Proceedings of the IEEE Nationale Aerospace and Electronics Conference.
Pengyu Zhou, **Daniel Chong**, Honglu Jiang, Junjie Zhang, Rui Dai. July 2025.

"An Empirical Study of Software Sanitization Locality." Proceedings of the IEEE National Aerospace and Electronics Conference. Nathaniel Boland, Junjie Zhang, **Daniel Chong**, and Rui Dai. July 2024.

"Automatically Inferring Image Base Addresses of ARM32 Binaries Using Architecture Features." Proceedings of the 12th International Symposium on Security and Privacy On Internet Of Things. **Daniel Chong**, Junjie Zhang, Nathaniel Boland, and Lingwei Chen. November 2023.

PRESENTATIONS

Daniel Chong. "Automatically Inferring Image Base Addresses of ARM32 Binaries Using Architecture Features." 12th International Symposium on Security and Privacy on Internet of Things 2024.

Daniel Chong. "Reverse Engineering ARM32." Great Lakes Region Aerospace Education Conference 2023.

VULNERABILITIES REPORTED IN OPEN-SOURCE SOFTWARE

PHP (100+), CPython (10), CRuby (8), NASA CryptoLib (2)

TECHNICAL SKILLS

- **Languages:** Python, PHP, JavaScript, C/C++
- **Vulnerability Research and Fuzzing**: AFL/AFL++, interpreter fuzzing, embedded systems fuzzing, LLM-driven fuzzing, IR-based fuzzing, constraint-guided seed synthesis.
- **Reverse Engineering and Binary Analysis:** Ghidra, Radare2, IDA, Unicorn, Capstone, symbolic execution, taint analysis, binary emulation.
- **ML & LLMS:** fine-tuning, prompt engineering, LLM-guided program transformation
- **Compiler Infrastructure:** LLVM, program mutation, IR design
- **Architectures:** ARM32, ARM64, x86-64, MIPS.